College of Information Science and Engineering, Huaqiao University
No. 668 Jimei Avenue, Xiamen, Fujian 361021, P. R. China Corresponding Author: Fang Duan, Ph.D. (nkfetsh@gmail.com)

Date: October 26, 2026 **Topic / Section:** Brain–Computer Interfaces & Neural Signal Processing

To: Prof. Warren M. Grill, Editor-in-Chief, *Journal of Neural Engineering* (JNE)
Department of Biomedical Engineering, Duke University, Durham, NC 27708, USA

Subject: Submission of Research Paper: **"Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Brain–Computer Interfaces"**

Dear Prof. Grill and Members of the Editorial Board,

We are pleased to submit our original research manuscript entitled **"Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Brain–Computer Interfaces"** for consideration for publication as a **Research Paper** in the *Journal of Neural Engineering* (JNE).

Context & Neural Engineering Challenge: Hybrid EEG–fNIRS interfaces offer a powerful non-invasive window into neural function by pairing high-temporal-resolution electrophysiology with localized capillary hemodynamics. However, developing zero-calibration, cross-subject brain–computer interfaces (BCIs) remains constrained by the fundamental biological timescale disparity between electrophysiological transients (milliseconds) and capillary hemodynamic responses (seconds). When unconstrained deep cross-attention networks are trained in cross-subject leave-one-subject-out (LOSO) regimes, they frequently learn spurious, acausal alignments where sluggish hemodynamic tokens mistakenly attend to future neural activations ($\delta_{ij} < 0$), impairing generalization to unseen participants.

Key Contributions & Empirical Rigor: To resolve this biophysical bottleneck, we propose **NVC-DelayFormer**, an architecture that embeds biophysical neurovascular timing priors directly into multimodal cross-attention. Our contributions are substantiated through rigorous participant-isolated evaluations and open-science benchmarking:

1. **Physiology-Guided Cross-Attention Formulation:** Multi-head cross-attention is parameterized by multicenter Gaussian delay priors grounded in canonical hemodynamic response delay windows ($\tau_h \in \{3.5, 4.5, 5.5, 6.5\}$ s) and token timestamp masks that explicitly eliminate acausal backwards-in-time connections, ensuring temporal precedence across modalities.
2. **Factorial Ablation & Dual-Cohort LOSO Evaluation:** In the primary 29-participant TU Berlin cohort, factorial ablations isolate the contributions of the delay prior and causal mask. Separately trained 15-fold LOSO evaluation on the independent Buccino et al. cohort, acquired with different EEG–fNIRS hardware, yielded 79.83% accuracy (Supplementary Section S.V).
3. **Unsupervised Recursive Covariance Alignment (CrEA):** To tackle inter-subject covariance drift during test time, we introduce Causal Recursive Euclidean Alignment (CrEA), which recursively updates reference covariances online without session lookahead, reducing relative covariance error from 45.15% to 29.38% within the first 8 trials. Across 29 participants and three random seeds, NVC-DelayFormer achieves a grand mean accuracy of **78.22% ± 1.00%** (primary seed: **79.37% ± 7.73%**), significantly exceeding standard 2D-EEGNet (72.89%, $p = 1.10 \times 10^{-4}$) and unconstrained cross-attention ($p = 7.18 \times 10^{-4}$). In task-specific training, accuracy reaches **73.24%** for working memory and **66.32%** for motor imagery.
4. **Compact Model and Measured Inference:** With **63,932 trainable parameters** and **24.79 M MACs**, the model requires 17.5× fewer parameters than EEG Conformer. Batch-one forward latency was **2.22 ms** on an NVIDIA RTX 4060 Laptop GPU.

Mandatory Declarations & Guidelines Compliance:

- **Originality & Exclusivity:** This manuscript represents original scientific work not published previously and not currently under consideration elsewhere.
- **Preprint & Conference Disclosure:** No preprint or prior conference version of this work exists, and no text recycling has occurred.
- **Authorship Approval:** All authors have read, critically reviewed, and approved the final manuscript and agree with its submission to JNE.
- **Ethical Approval:** This study reanalyzed de-identified public data. The original TU Berlin collections were approved under protocol SH_01_20150330; the Buccino et al. collection was approved by the University of Houston Institutional Review Board. Both original studies reported written participant consent.
- **Conflict of Interest & Funding:** The authors declare no competing financial or non-financial interests. No specific external grant supported this research.
- **Reproducibility & Open Science:** Source code, available example checkpoints, test suites, and audit logs are openly accessible at GitHub (ujujuj3061831150/nvc-delayformer-bci).

We believe that the interdisciplinary neuroengineering methodology and rigorous dual-cohort validation presented in this paper will be of strong interest to the readership of the *Journal of Neural Engineering*. Should the Editor or reviewers require any further details, clarifications, or access to intermediate audit artifacts, please do not hesitate to contact the corresponding author; we remain entirely at your disposal throughout the evaluation process.

Thank you very much for your time, consideration, and thoughtful evaluation of our work.

Respectfully submitted,

Fang Duan, Ph.D. (Corresponding Author)
Lecturer
College of Information Science and Engineering
Huaqiao University, Xiamen 361021, China
E-mail: nkfetsh@gmail.com
ORCID: 0000-0003-0982-2206

Hongjie Chen (First Author)
Graduate Researcher
College of Information Science and Engineering
Huaqiao University, Xiamen 361021, China
E-mail: chenhj976@gmail.com